

class10

Alishba Qadri

```
# download csv file  
read.csv("pdb_stats.csv")
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	214078			
2	1	0	13981			
3	0	0	15759			
4	3	1	4941			
5	0	0	237			
6	0	4	22			

Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

A1. The percentage of structures solved by X-Ray Microscopy are 80.95%. The percentage of structures solved by EM are 12.84%.

Total: 249018 X-Ray Total: 201582 EM Total: 31970

Q2. What proportion of structures in the PDB are protein?

A2. The proportion of structures that are protein are 97.91%.

$243818 / 249018 = 97.91\%$

Q3. Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

A3. There are 881 HIV-1 protease structures in the current PDB.

3. Visualizaing the HIV-1 protease structure



Q4. Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

A4. We only see one atom, mainly oxygen, per water molecule, because it contains much more electrons than hydrogen. Therefore, oxygen shows up much more clearer than hydrogen, as the software tracks by electron density.

Q5. There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

A5. Yes, this water molecule has a residue number of HOH308. It is situated in the middle of the polymer and the two HIV-1 protease chains.

Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



4. Introduction to Bio3D in R

```
# load bio3d
library(bio3d)

# read pdb file
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7. How many amino acid residues are there in this pdb object?

A7. There are 198 amino acid residues

Q8. Name one of the two non-protein residues.

A8. One of the two non-protein residues is water (HOH).

Q9. How many protein chains are in this structure?

A9. There are 2 chains in this structure.

```
# find attributes of the pdb file
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

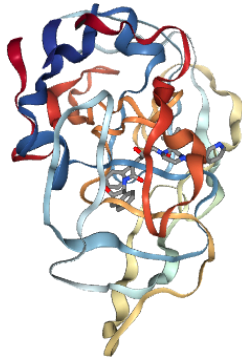
```
# access the atom attribute
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

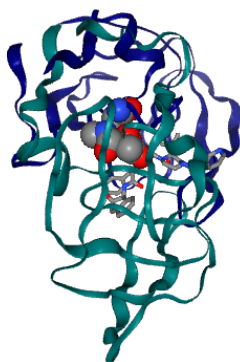
```
# load packages
library(bio3dview)
library(NGLViewerR)

# generate a NGL structure overview of bio3d pdb
view.pdb(pdb) |>
  setSpin()
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
          highlight = sele,
          highlight.style = "spacefill") |>
setRock()
```



```
# read adk (Adenylate Kinase) file
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

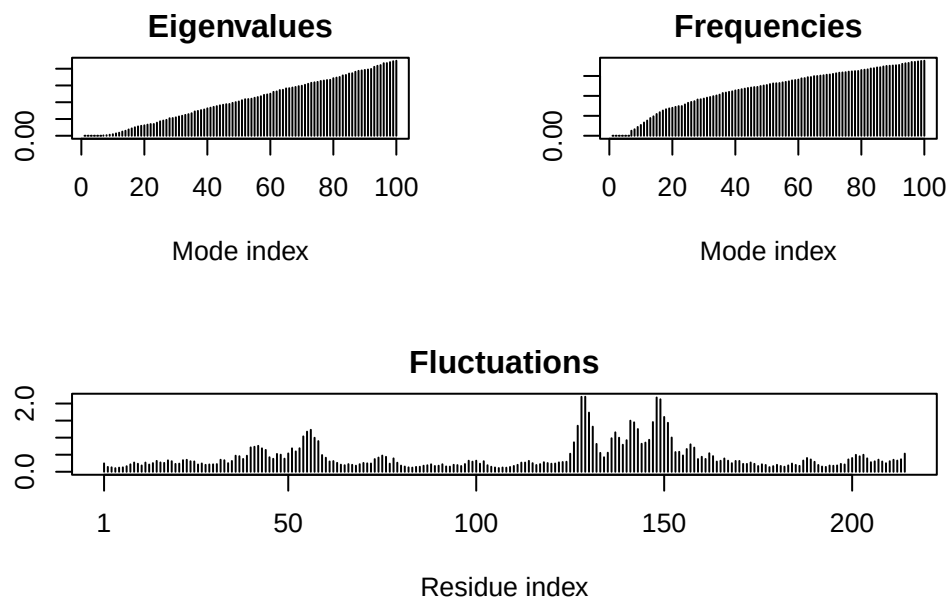
Protein sequence:
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

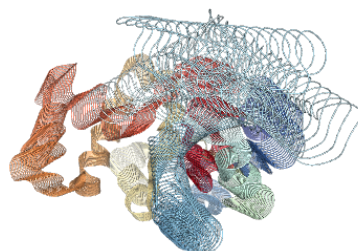
```
# Perform flexibility prediction
m <- nma(adk)
```

Building Hessian... Done in 0.01 seconds.
Diagonalizing Hessian... Done in 0.181 seconds.

```
# create plot
plot(m)
```



```
# view display using view.nma() function  
view.nma(m, pdb=adk)
```



5. Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

A10. The msa package is found only on BioConductor, due to the BiocManager::install() function.

Q11. Which of the above packages is not found on BioConductor or CRAN?

A11. The bioboot/bio3dview package is not found on BioConductor or CRAN, and needs to be installed using the devtools::install_github() function.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

A12. True.

```
# load package
library(bio3d)

# view query sequence for chain A
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRPTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
```

```

      121      .      .      .      .      .      .      180
      181      .      .      .      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      .      .      .      214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

A13. There are 214 amino acids in this sequence.

```
# use sequence as a query to find similar sequences on PDB
hits <- NULL
hits$ pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/3HPR.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

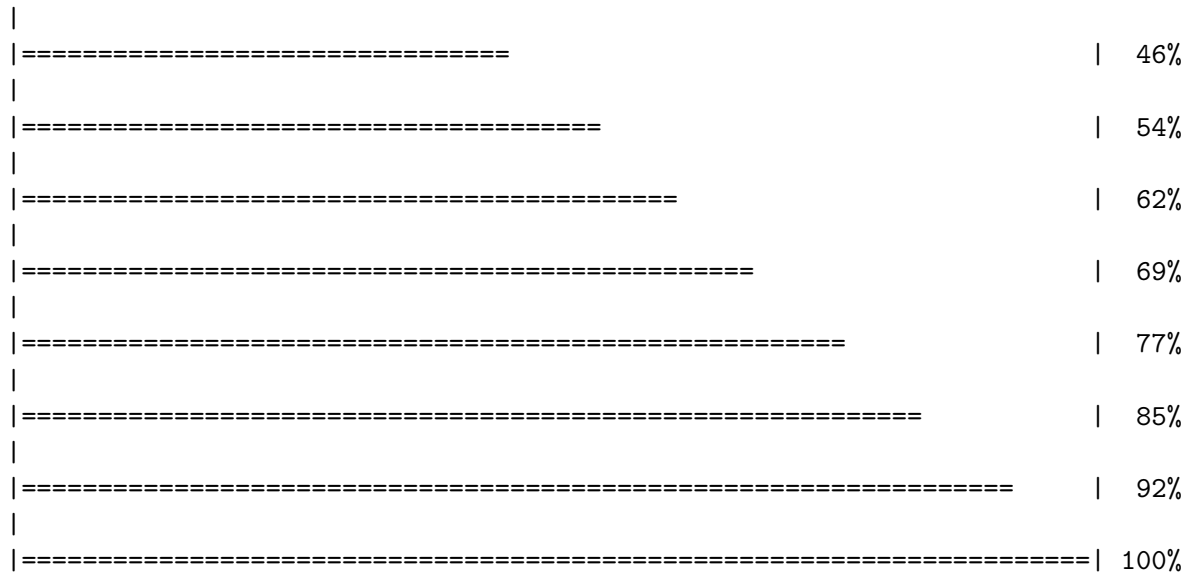
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download

	0%
=====	8%
=====	15%
=====	23%
=====	31%
=====	38%



```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

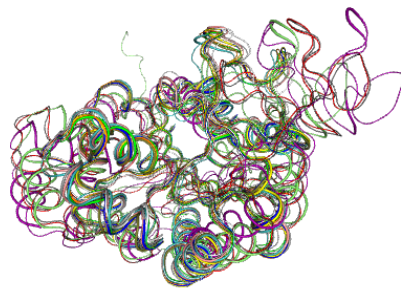
```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

```
pdb/seq: 1    name: pdbs/split_chain/1AKE_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2    name: pdbs/split_chain/6S36_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3    name: pdbs/split_chain/6RZE_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4    name: pdbs/split_chain/3HPR_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5    name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6    name: pdbs/split_chain/5EJE_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7    name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8    name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9    name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb
             PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

```
# view results with bio3dview
library(bio3dview)

view.pdbs(pdbs)
```



```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
# view annotation data
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					

4PZL_A	4PZL	A	Protein	242	X-ray
	resolution	scopDomain			pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
3X2S_A	2.80	<NA>		<NA>	
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)		
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3GMT_A	2.10	<NA>		<NA>	
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)		
	ligandId				
1AKE_A	AP5				
6S36_A	CL (3),NA,MG (2)				
6RZE_A	NA (3),CL (2)				
3HPR_A	AP5				
1E4V_A	AP5				
5EJE_A	AP5,C0				
1E4Y_A	AP5				
3X2S_A	JPY (2),AP5,MG				
6HAP_A	AP5				
6HAM_A	AP5				
4K46_A	ADP,AMP,PO4				
3GMT_A	SO4 (2)				
4PZL_A	CA,FMT,GOL				
					ligandName
1AKE_A					BIS(ADENOSINE)-5'-
PENTAPHOSPHATE					
6S36_A					CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A					SODIUM ION (3),CHLORIDE ION (2)
3HPR_A					BIS(ADENOSINE)-5'-
PENTAPHOSPHATE					
1E4V_A					BIS(ADENOSINE)-5'-
PENTAPHOSPHATE					
5EJE_A					BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A					BIS(ADENOSINE)-5'-
PENTAPHOSPHATE					
3X2S_A					N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION

6HAP_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6HAM_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A	SULFATE ION (2)
4PZL_A	CALCIUM ION,FORMIC ACID,GLYCEROL

	source
1AKE_A	Escherichia coli
6S36_A	Escherichia coli
6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
3GMT_A	Burkholderia pseudomallei 1710b
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBIT

6S36_A
6RZE_A
3HPR_A
1E4V_A

loop

5EJE_A

1E4Y_A

loop

3X2S_A

conjugated adenylate kinase

6HAP_A

6HAM_A

4K46_A

3GMT_A

4PZL_A

Cryst

The crys

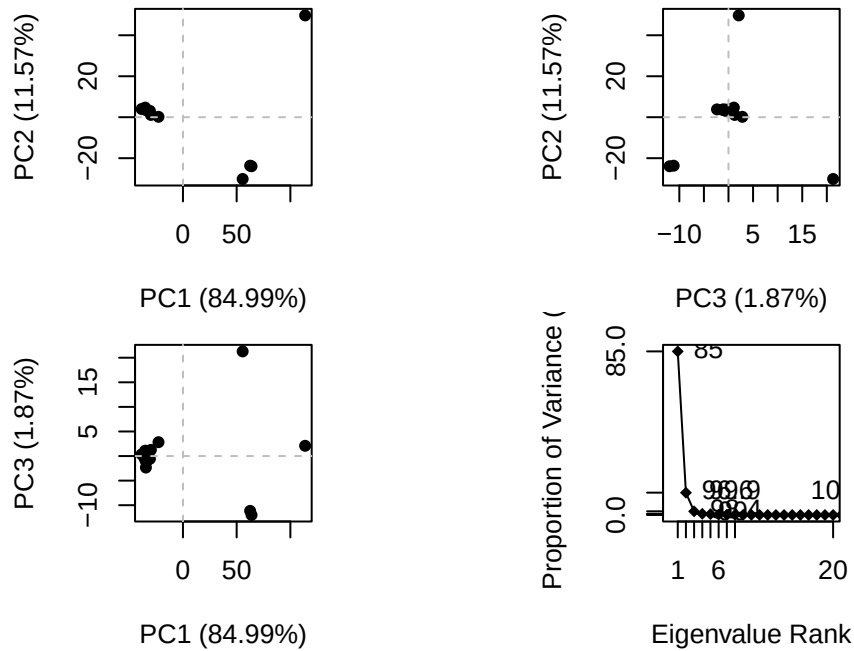
		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA	
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560	
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500	
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320	

1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680

rWork spaceGroup

1AKE_A	0.19600	P	21	2	21
6S36_A	0.15940	C	1	2	1
6RZE_A	0.18190	C	1	2	1
3HPR_A	0.20620	P	21	21	2
1E4V_A	0.19600	P	21	2	21
5EJE_A	0.18630	P	21	2	21
1E4Y_A	0.17800	P	1	21	1
3X2S_A	0.20700	P	21	21	21
6HAP_A	0.22370	I	2	2	2
6HAM_A	0.20311	P	43		
4K46_A	0.16730	P	21	21	21
3GMT_A	0.23500	P	1	21	1
4PZL_A	0.19130	P	32		

```
# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)
```

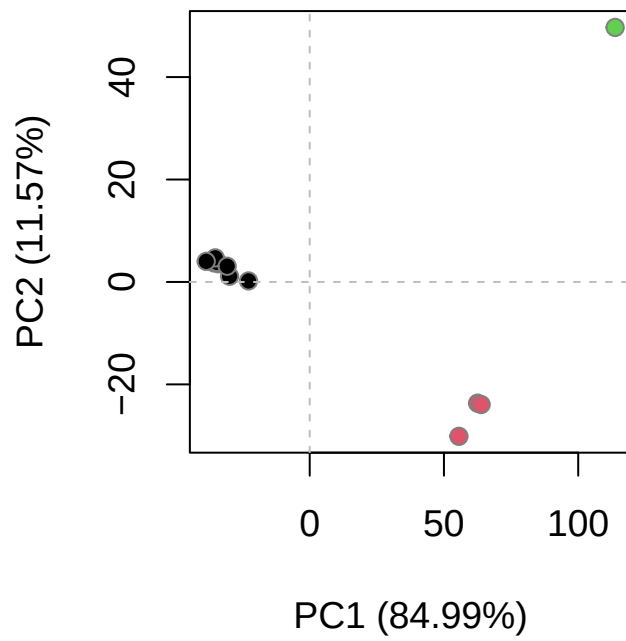


```
# Calculate RMSD
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

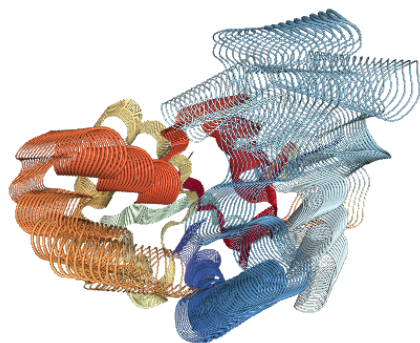
```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
# Visualize first principal component  
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
# view results  
view.pca(pc.xray)
```



```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                  PC2=pc.xray$z[,2],
                  col=as.factor(grps.rd),
                  ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```

